The Mathematical Tools of Quantum Mechanics (Cohen–Tannoudji, Ch. II) Section E. Two Important Examples of Representations and Observables

Beatriz Gonzalez del Rio

Outline

- A. One-particle wave function space
- B. State space and Dirac notation
- C. Representations in the state space
- D. Eigenvalue equations and observables
- E. Two key examples: $|\boldsymbol{r}\rangle$ and $|\boldsymbol{p}\rangle$; operators $\boldsymbol{R}$, $\boldsymbol{P}$
- F. Tensor products and applications

Wave functions and kets

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- Wave functions $\psi(\mathbf{r}) \leftrightarrow$ vectors (kets) $|\psi\rangle$.
- The scalar product in $\mathcal{E}_r$ coincides with the usual L^2 scalar product.

$|\mathbf{r}\rangle$ and $|\mathbf{p}\rangle$ representations

We introduce two particular bases of F:

$$\xi_{\mathbf{r}_0}(\mathbf{r}) = \delta(\mathbf{r} - \mathbf{r}_0), \qquad v_{\mathbf{p}_0}(\mathbf{r}) = (2\pi\hbar)^{-3/2}\, e^{\frac{i}{\hbar}\mathbf{p}_0\cdot\mathbf{r}}.$$

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- $|\mathbf{r}_0\rangle$: "position" basis.
- $|\mathbf{p}_0\rangle$: "momentum" basis.
- Indices $\mathbf{r}_0$, $\mathbf{p}_0$ are continuous (e.g. $x_0, y_0.z_0$).

Orthonormalization and closure relations

The bases $\{|\mathbf{r}_0\rangle\}$ and $\{|\mathbf{p}_0\rangle\}$ are orthonormal (in the generalized sense):

$$\langle \mathbf{r}_0 | \mathbf{r}_0' \rangle = \delta(\mathbf{r}_0 - \mathbf{r}_0'), \qquad \langle \mathbf{p}_0 | \mathbf{p}_0' \rangle = \delta(\mathbf{p}_0 - \mathbf{p}_0').$$

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They satisfy the closure (completeness) relations:

$$\int d^3 r_0 \; |\mathbf{r}_0\rangle \langle \mathbf{r}_0| = \mathbb{1}, \qquad \int d^3 p_0 \; |\mathbf{p}_0\rangle \langle \mathbf{p}_0| = \mathbb{1}.$$

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Interpretation:

- The sets $\{|\mathbf{r}_0\rangle\}$ and $\{|\mathbf{p}_0\rangle\}$ each form a basis of $\mathcal{E}_r$.
- The closure operators above are the identity operator.

Components of a ket

For an arbitrary ket $|\psi\rangle$, the closure relations give two equivalent expansions:

$$|\psi\rangle = \int d^3 r_0 \; |\mathbf{r}_0\rangle \langle \mathbf{r}_0 | \psi \rangle \,, \qquad |\psi\rangle = \int d^3 p_0 \; |\mathbf{p}_0\rangle \langle \mathbf{p}_0 | \psi \rangle \,.$$

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The components in these bases are:

$$\langle \mathbf{r}_0 | \psi \rangle = \psi(\mathbf{r}_0), \qquad \langle \mathbf{p}_0 | \psi \rangle = \widetilde{\psi}(\mathbf{p}_0),$$

where $\widetilde{\psi}(\mathbf{p})$ is the Fourier transform of $\psi(\mathbf{r})$.

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Comments:

- $\psi(\mathbf{r})$: wave function in the position representation.
- $\tilde{\psi}(\mathbf{p})$: wave function in the momentum representation.

$|\mathbf{r}\rangle$, $|\mathbf{p}\rangle$ and basic relations

We now write the basis kets simply as $|\mathbf{r}\rangle$ and $|\mathbf{p}\rangle$. Then the components become

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The orthonormalization and closure relations read:

$$\langle \mathbf{r}|\mathbf{r}'\rangle = \delta(\mathbf{r}-\mathbf{r}'), \qquad \int d^3r \ |\mathbf{r}\rangle \langle \mathbf{r}| = \mathbb{1},$$

$$\langle \mathbf{p}|\mathbf{p}'\rangle = \delta(\mathbf{p}-\mathbf{p}'), \qquad \int d^3p \ |\mathbf{p}\rangle \langle \mathbf{p}| = \mathbb{1}.$$

Example: discrete orthonormal basis $\{|u_i\rangle\}$

Let $\{|u_i\rangle\}$ be an orthonormal basis of $\mathcal{E}_r$, with

$$\sum_i |u_i\rangle \langle u_i| = \mathbb{1}.$$

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Define the wavefunctions

$$u_i(\mathbf{r}) = \langle \mathbf{r}|u_i\rangle .$$

Evaluating the closure relation between $\langle \mathbf{r}|$ and $|\mathbf{r}'\rangle$ gives

$$\sum_i \langle \mathbf{r}|u_i\rangle \langle u_i|\mathbf{r}'\rangle = \langle \mathbf{r}|\mathbf{r}'\rangle = \sum_i u_i(\mathbf{r})\, u_i^*(\mathbf{r}') = \delta(\mathbf{r}-\mathbf{r}').$$

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Comment:

- This is the closure relation expressed entirely in the $|\mathbf{r}\rangle$ representation.

Scalar product in position and momentum representations

In the $|\mathbf{r}\rangle$ representation:

$$\langle\phi|\psi\rangle = \int d^3r \; \langle\phi|\mathbf{r}\rangle \, \langle\mathbf{r}|\psi\rangle = \int d^3r \, \phi^*(\mathbf{r}) \, \psi(\mathbf{r}).$$

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$$\langle\phi|\psi\rangle = \int d^3p \; \langle\phi|\mathbf{p}\rangle \, \langle\mathbf{p}|\psi\rangle = \int d^3p \, \widetilde{\phi}^*(\mathbf{p}) \, \widetilde{\psi}(\mathbf{p}).$$

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Interpretation:

- The scalar product is independent of the representation.
- Wavefunctions in r-space and p-space carry exactly the same physical information.

Change from $|\mathbf{r}\rangle$ to $|\mathbf{p}\rangle$

The change of representation is governed by

$$\langle \mathbf{r}|\mathbf{p}\rangle = \langle \mathbf{p}|\mathbf{r}\rangle^* = (2\pi\hbar)^{-3/2}\, e^{\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{r}}.$$

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From $\tilde{\psi}(\mathbf{p}) = \langle \mathbf{p}|\psi\rangle$, we obtain the position-space wavefunction:

$$\psi(\mathbf{r}) = \langle \mathbf{r}|\psi\rangle = \int d^3p\ \langle \mathbf{r}|\mathbf{p}\rangle\, \langle \mathbf{p}|\psi\rangle = (2\pi\hbar)^{-3/2} \int d^3p\, e^{\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{r}}\, \tilde{\psi}(\mathbf{p}).$$

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Comment: this is just the Fourier transform from momentum space to position space.

Inverse transform and operator matrix elements

Conversely, using $\langle \mathbf{p}|\mathbf{r}\rangle = \langle \mathbf{r}|\mathbf{p}\rangle^*$, we get

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For an operator A, its matrix elements in the two representations are related by

$$A(\mathbf{p}', \mathbf{p}) = \langle \mathbf{p}'|\mathbf{r}'\rangle \, \langle \mathbf{r}'| \, A \, |\mathbf{r}\rangle \, \langle \mathbf{r}|\mathbf{p}\rangle = (2\pi\hbar)^{-3} \int d^3r \, d^3r' \; e^{-\frac{i}{\hbar}\mathbf{p}'\cdot\mathbf{r}'} \, A(\mathbf{r}', \mathbf{r}) \, e^{\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{r}}.$$

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Comments:

- $\langle \mathbf{r}|\mathbf{p}\rangle$ connects all r- and p-space quantities.
- Same operator A, different matrix representations.

The X operator in the $|\mathbf{r}\rangle$ representation

Let $|\psi\rangle \in \mathcal{E}_r$ and denote the corresponding wave function by

$$\psi(\mathbf{r}) = \langle \mathbf{r}|\psi\rangle = \psi(x, y, z).$$

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$$|\psi'\rangle = X\,|\psi\rangle \qquad \Longrightarrow \qquad \psi'(\mathbf{r}) = \langle \mathbf{r}|X|\psi\rangle\,.$$

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In the $|\mathbf{r}\rangle$ representation,

$$\psi'(\mathbf{r}) = x\,\psi(\mathbf{r}),$$

so X acts as multiplication by x. (Acts on state space)

X, Y, Z operators

The three components of the position operator $\mathbf{R}$ are defined by:

$$\langle \mathbf{r}|X|\psi\rangle = x\ \langle \mathbf{r}|\psi\rangle\,, \qquad \langle \mathbf{r}|Y|\psi\rangle = y\ \langle \mathbf{r}|\psi\rangle\,,$$

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The numbers x, y, z are the three indices labeling $|\mathbf{r}\rangle$. Think:
$\mathbf{r} = (x, y, z)$ and $X, Y, Z =$ components of one vector operator.

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Matrix element $\langle\phi|\, X\, |\psi\rangle$:

$$\langle\phi|\, X\, |\psi\rangle = \int \mathrm{d}^3 r\ \langle\phi|\mathbf{r}\rangle\, \langle \mathbf{r}|\, X\, |\psi\rangle = \int \mathrm{d}^3 r\, \phi^*(\mathbf{r})\, x\, \psi(\mathbf{r})$$

Definition of the momentum operators

Similarly, the components of the momentum operator $\mathbf{P}$ satisfy in the $|\mathbf{p}\rangle$ representation:

$$\langle \mathbf{p}|P_x|\psi\rangle = p_x \, \langle \mathbf{p}|\psi\rangle \,, \qquad \langle \mathbf{p}|P_y|\psi\rangle = p_y \, \langle \mathbf{p}|\psi\rangle \,,$$

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- P_x multiplies by p_x in momentum space.
- Indices p_x, p_y, p_z label $|\mathbf{p}\rangle$.

Action of $\mathbf{P}$ in the $|\mathbf{r}\rangle$ representation

Insert closure and transform:

$$\langle \mathbf{r}|P_x|\psi\rangle = \int d^3p\ \langle \mathbf{r}|\mathbf{p}\rangle \langle \mathbf{p}|\ P_x\ |\psi\rangle = (2\pi\hbar)^{-3/2}\int d^3p\ p_x\ e^{\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{r}}\ \tilde{\psi}(\mathbf{p}).$$

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Now use the key identity:

$$\frac{\partial}{\partial x}e^{ipr/\hbar} = \frac{ip_x}{\hbar}e^{ipr/\hbar} \quad \Rightarrow \quad p_x\, e^{ipr/\hbar} = -i\hbar\,\frac{\partial}{\partial x}e^{ipr/\hbar}.$$

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Therefore,

$$p_x\tilde{\psi}(\mathbf{p})e^{ipr/\hbar} = -i\hbar\tilde{\psi}(\mathbf{p})\frac{\partial}{\partial x}e^{ipr/\hbar}$$

The factor p_x becomes a derivative acting on the exponential.

Action of $\mathbf{P}$ in the $|\mathbf{r}\rangle$ representation

Recognizing the Fourier transform of $\psi(\mathbf{r})$ gives the key result:

$$\begin{aligned}\langle \mathbf{r}|P_x|\psi\rangle =&(2\pi\hbar)^{-3/2}\int d^3p\,(-i\hbar\frac{\partial}{\partial x}\tilde{\psi}(\mathbf{p})e^{ipr/\hbar}) = \\ =&-i\hbar\frac{\partial}{\partial x}((2\pi\hbar)^{-3/2}\int d^3p\,\tilde{\psi}(\mathbf{p})e^{ipr/\hbar}) = -i\hbar\,\frac{\partial}{\partial x}\psi(\mathbf{r})\end{aligned}$$

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Thus, in the $|\mathbf{r}\rangle$ representation:

$$\mathbf{P} = -i\hbar\,\nabla.$$

Canonical commutation relations

Using the action of X and P_x on wave functions:

$$\langle \mathbf{r}| \left[X, P_x\right] |\psi\rangle = \langle \mathbf{r}| \left(XP_x - P_x X\right) |\psi\rangle = x \langle \mathbf{r}| P_x |\psi\rangle + i\hbar \frac{\partial}{\partial x} \langle \mathbf{r}| X |\psi\rangle =$$

$$= -i\hbar x \frac{\partial}{\partial x} \langle \mathbf{r}|\psi\rangle + i\hbar \frac{\partial}{\partial x} x \langle \mathbf{r}|\psi\rangle = i\hbar \langle \mathbf{r}|\psi\rangle$$

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$$= -i\hbar x \frac{\partial}{\partial x} \langle \mathbf{r}|\psi\rangle + i\hbar \frac{\partial}{\partial x} x \langle \mathbf{r}|\psi\rangle = i\hbar \langle \mathbf{r}|\psi\rangle$$

Therefore

$$[X, P_x] = i\hbar \mathbb{1}.$$

Similarly, the full set of canonical commutators is:

$$[R_i, R_j] = 0, \qquad [P_i, P_j] = 0, \qquad [R_i, P_j] = i\hbar\, \delta_{ij}.$$

Hermiticity of R and P

For X:

$$\langle\phi|\, X\, |\psi\rangle = \int d^3r\, \phi^*(\mathbf{r})\, x\, \psi(\mathbf{r}) = \left(\int d^3r\, \psi^*(\mathbf{r})\, x\, \phi(\mathbf{r})\right)^* = \langle\psi|\, X\, |\phi\rangle^*\,,$$

so $X = X^\dagger$.

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so $X = X^\dagger$.

For P_x, integration by parts yields:

$$\langle\phi|\, P_x\, |\psi\rangle = -\, i\hbar \int dy\, dz \int dx\, \phi^*(\mathbf{r}) \frac{\partial}{\partial x}\psi(\mathbf{r}) = i\hbar \int d^3r \psi(\mathbf{r}) \frac{\partial}{\partial x}\phi^*(\mathbf{r}) =$$

$$= \left[-i\hbar \int d^3r \psi^*(\mathbf{r}) \frac{\partial}{\partial x}\phi(\mathbf{r})\right]^* = \langle\psi|\, P_x\, |\phi\rangle^*$$

Thus $P_x = P_x^\dagger$, and analogously for P_y, P_z.

Hermiticity of R and P

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"$\frac{\partial}{\partial x}$ is not Hermitian, but $-i\hbar\frac{\partial}{\partial x}$ **is** Hermitian."

Eigenvectors of the position operator

From

$$\langle \mathbf{r}|X|\mathbf{r}_0\rangle = x\,\langle \mathbf{r}|\mathbf{r_0}\rangle = x\,\delta(\mathbf{r}-\mathbf{r}_0) = x_0\,\delta(\mathbf{r}-\mathbf{r}_0) = x_0\,\langle \mathbf{r}|\mathbf{r_0}\rangle$$

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Thus $|\mathbf{r}\rangle$ are simultaneous eigenkets of X, Y, Z.

Eigenvectors of the momentum operator

From the action of P_x in $|\mathbf{p}\rangle$:

$$P_x |\mathbf{p}\rangle = p_x |\mathbf{p}\rangle .$$

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Using $\langle \mathbf{r}|\mathbf{p}\rangle = (2\pi\hbar)^{-3/2} e^{\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{r}}$, we obtain the components:

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Conclusion: $|\mathbf{p}\rangle$ are simultaneous eigenvectors of P_x, P_y, P_z.

R and P as observables

The position basis satisfies:

$$\int d^3r \ |\mathbf{r}\rangle \langle \mathbf{r}| = \mathbb{1}.$$

The momentum basis satisfies:

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Thus both sets form bases of $\mathcal{E}_r$, and therefore R and P are observables.

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Thus both sets form bases of $\mathcal{E}_r$, and therefore R and P are observables. Additionally, the three eigenvalues x_0, y_0, z_0 uniquely determines the eigenvector $|\mathbf{r_0}\rangle$. Therefore, the set of the three operators X, Y, Z constitutes a C.S.C.O. in $\mathcal{E}_r$. Same with P_x, P_y, P_z.

Complete sets of commuting observables (CSCO)

Since $[R_i, R_j] = 0$ and $[P_i, P_j] = 0$, the sets

$$\{X, Y, Z\}, \qquad \{P_x, P_y, P_z\}$$

are each CSCOs.

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are each CSCOs.

However, in $\mathcal{E}_r$, X by itself is NOT a C.S.C.O. x_0 is fixed, but y_0 and z_0 can take any value.

But, in one dimension ($\mathcal{E}_x$), the position operator alone forms a C.S.C.O.:

$$X \quad \text{determines uniquely the eigenket} \quad |x_0\rangle .$$

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- Two C.S.C.O.s found in $\mathcal{E}_r$: X, Y, Z and P_x, P_y, P_z.